Statistical Learning and Data Analysis Module B

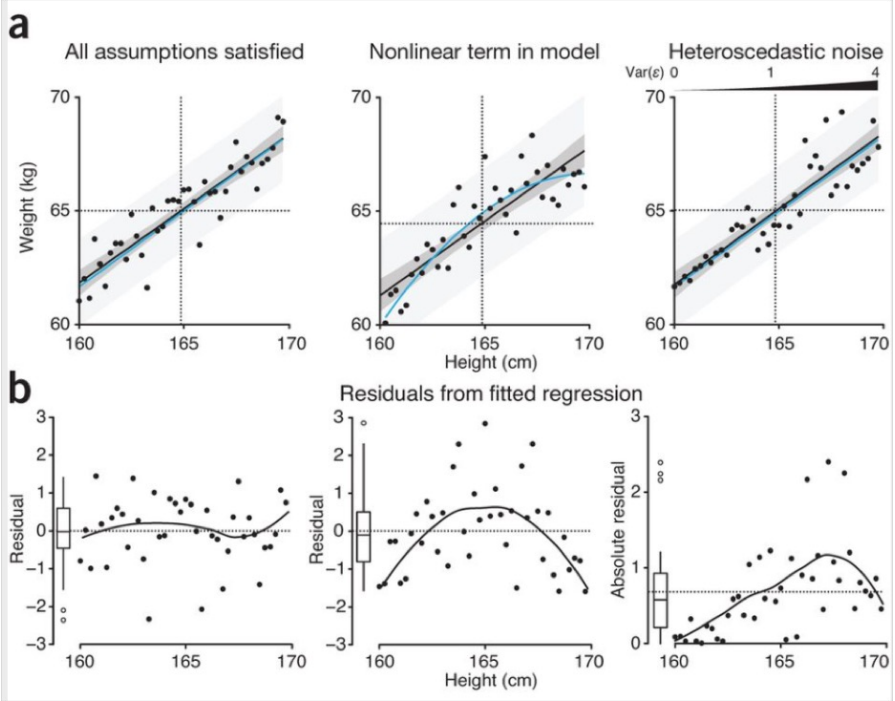
Regression Diagnostics

Roberta Siciliano

Statistics, tèchnè-loghìa, analysis of data

DIETI, University of Napoli Federico II

Regression diagnostics



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Potential problems

Model mis-specification and Non-linearity

Correlation of error terms

Non-constant variance of error terms

Outliers

High-leverage points

Collinearity

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Regression diagnostics: why and how

If any of the assumptions is violated (i.e., if there is nonlinearity, serial correlation, heteroscedasticity, and/or non-normality), then the forecasts, confidence intervals, and any insights yielded by a regression model may be (*at best*) **inefficient** or (*at worst*) seriously **biased or misleading**.

Regression diagnostics include **statistical tests** and **graphical analysis of residuals**.

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About diagnostic testing procedures

Aim is to check the consistency of the assumptions of the regression model with the data.

The diagnostic tests, or **tests of incorrect specification**, testing the classic hypotheses of the model, have the purpose of checking the efficiency and correctness of the estimate of the parameters with the OLS.

Any refusal of the null hypothesis of a diagnostic test does not have the consequence of considering the alternative hypothesis as valid, but only leads to an alarm bell of inconsistency between what is hypothesized and the available observations.

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The logic of diagnostic tests

They are based on the analysis of the residuals of the OLS and are often calculated as F statistics, by comparing RSS_c of the (constrained) fitted model with RSS of a more general model.

In such contexts, the unconstrained model is called the **auxiliary model**.

Diagnostic tests are interpreted as *problems of omitted variables*.

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Diagnostic Test Specification

The Model: $\mathbf{y} = \mathbf{X}\beta + \boldsymbol{\varepsilon}$

The Auxiliary Model: $\mathbf{y} = \mathbf{X}\beta^* + \mathbf{W}\gamma + \eta$

Including h omitted predictors (auxiliary variables)

If the auxiliary model is correct (thus $\gamma \neq 0$), then the OLS estimates of the fitted model are biased

$$E\left(\hat{\beta}|\mathbf{X},\mathbf{W}\right) = E[(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T(\mathbf{X}\beta^* + \mathbf{W}\gamma + \eta\,)|\mathbf{X},\mathbf{W}]$$

$$= \beta^* + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}\gamma + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T E(\eta|\mathbf{X},\mathbf{W})$$

$$= \beta^* + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\,\mathbf{W}\gamma \neq \beta$$

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RESET (Regression Specification Error Test) of Ramsey

The auxiliary model is more general.

The current model is one constrained version.

If the correct model is the auxiliary model, the residuals are

$$e = \mathbf{y} - \mathbf{X}\hat{\beta} = \mathbf{X}\beta^* + \mathbf{W}\gamma + \eta - \mathbf{X}\hat{\beta} = \mathbf{X}\left(\beta^* - \hat{\beta}\right) + \mathbf{W}\gamma + \eta$$

If $\gamma = 0$ then the Residual Sum of Squares are equivalent in the two above models so that we can use a F-statistic:

$$F = \frac{(RSS_c - RSS)/h}{RSS/(n-p-h-1)} \approx F_{(h),(n-p-h-1)}$$

where RSS_c is the Residual Sum of Squares of the (constrained) fitted model whereas RSS is the Residual Sum of Squares of the more general model.

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When to apply RESET

Problems:

- If $h > (n - p)$ then the test cannot be computed
- The auxiliary variables are not known
- The auxiliary variables are considered as proxy, such to understand potential un-correct specification errors

The RESET test is useful because it signals something wrong with the specification:

✓**The inappropriate choice of the functional relationship**

✓**The omission of important variables**

The RESET test does not indicate specific alternatives to the given model. It is said to be a "non-constructive test" as it gives *no indication of how to act to overcome the problem.*

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How to apply RESET

Adding the powers of the fitted response as auxiliary variables:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \gamma_1 \hat{Y}^2 + \gamma_2 \hat{Y}^3 + \ldots + \gamma_h \hat{Y}^{(h+1)} + \eta$$

We can test the null hypothesis that $\gamma_j = 0$ for all $j = 1, \ldots, h$ in order to understand if the systematic component of the model has some important missingness.

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```
7. Attribute Information:
    1. CRIM      per capita crime rate by town
    2. ZN        proportion of residential land zoned for lots over
                 25,000 sq.ft.
    3. INDUS     proportion of non-retail business acres per town
    4. CHAS      Charles River dummy variable (= 1 if tract bounds
                 river; 0 otherwise)
    5. NOX       nitric oxides concentration (parts per 10 million)
    6. RM        average number of rooms per dwelling
    7. AGE       proportion of owner-occupied units built prior to 1940
    8. DIS       weighted distances to five Boston employment centres
    9. RAD       index of accessibility to radial highways
    10. TAX      full-value property-tax rate per $10,000
    11. PTRATIO  pupil-teacher ratio by town
    12. B        1000(Bk - 0.63)^2 where Bk is the proportion of blacks
                 by town
    13. LSTAT    % lower status of the population
    14. MEDV     Median value of owner-occupied homes in $1000's
```

Boston Housing Data – UCI Machine Learning Repository
(http://archive.ics.uci.edu/ml/)

Target is the Median Value of owner-occupied homes in $1000's

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RESET identifies a mis-specification.

The RESET test does not indicate specific alternatives to the given model.

It is said to be a "non-constructive test" as it gives *no indication of how to act to overcome the problem.*

```
Regressione ausiliaria per il test RESET di specificazione
Stime OLS usando le 506 osservazioni 1-506
Variabile dipendente: MEDV

      VARIABILE      COEFFICIENTE   ERRORE STANDARD     T STAT    2Prob(t>|T|)

         const        10,9455            9,21242        1,188    0,235357
          CRIM        -0,0727011         0,0337855     -2,152    0,031896 **
            ZN        -0,0580182         0,0168952     -3,434    0,000645 ***
          CHAS        -1,34686           0,978111      -1,377    0,169138
          NOX_         7,11834           5,65912        1,258    0,209042
            RM        -2,33587           0,943615      -2,475    0,013643 **
           DIS         0,861378          0,411045       2,096    0,036630 **
           RAD        -0,0381349         0,0889514     -0,429    0,668318
          TAX_         0,00396045        0,00404091     0,980    0,327524
       PTRATIO         0,432274          0,267043       1,619    0,106143
             B         0,000165326       0,00313148     0,053    0,957917
         LSTAT        -0,139267          0,125903      -1,106    0,269204
        yhat^2         0,0125086         0,0111642      1,120    0,263084
        yhat^3         0,000343746       0,000166703    2,062    0,039730 **


Statistica test: F = 95,832152,
con p-value = P(F(2,492) > 95,8322) = 7,11e-036

Test RESET di specificazione -
  Ipotesi nulla: la specificazione è adeguata
  Statistica test: F(2, 492) = 95,8322
  con p-value = P(F(2, 492) > 95,8322) = 7,11313e-036
```

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Penn World data set in Gretl

```
Modello 1: Stime OLS usando le 50 osservazioni 1951-2000
Variabile dipendente: consumo

      VARIABILE      COEFFICIENTE   ERRORE STANDARD     T STAT    2Prob(t>|T|)

         const         51,7047            0,673070     76,819   < 0,00001 ***
           pil          0,000872364       6,05604E-05  14,405   < 0,00001 ***

  Media della variabile dipendente = 59,0282
  Deviazione standard della variabile dipendente = 7,12191
  Somma dei quadrati dei residui = 466,917
  Errore standard dei residui = 3,11888
  R-quadro = 0,812133
  R-quadro corretto = 0,808219
  Gradi di libertà = 48
  Statistica Durbin-Watson = 0,509841
  Coefficiente di autocorrelazione del prim'ordine = 0,763042
  Log-verosimiglianza = -126,8
  Criterio di informazione di Akaike (AIC) = 257,6
  Criterio bayesiano di Schwarz (BIC) = 261,424
```

```
Test Chow per break strutturale all'osservazione 1977 -
  Ipotesi nulla: nessun break strutturale
  Statistica test: F(2, 46) = 1,183
  con p-value = P(F(2, 46) > 1,183) = 0,315506
```

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Non-Linearity of the Data

How to detect: nonlinearity is usually most evident in **a plot of the *observed versus predicted values*** *(the points should be symmetrically distributed around a diagonal line)* or **a plot of *residuals versus predicted values*** *(the points should be symmetrically distributed around a horizontal line)* which are a part of standard regression output. Look carefully for evidence of a "bowed" pattern, indicating that the model makes systematic errors whenever it is making unusually large or small predictions.

How to fix: consider applying a *nonlinear transformation* to the dependent and/or independent variables, namely log transformation or squared predictor, polynomial fit, etc.

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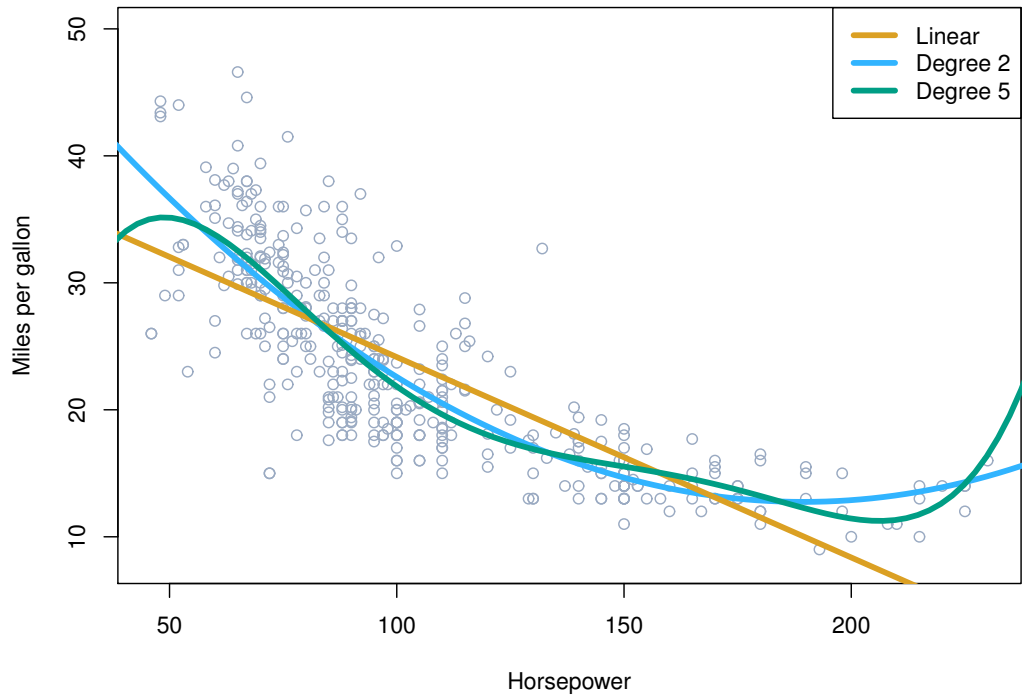
Non-Linear Relationships

Auto data set of a number of cars

Mpg (gas mileage in miles per gallon) *versus*

- ✓ **Horsepower**
- ✓ **Polynomial of Horsepower up to second-degree**
- ✓ **All polynomials of horsepower up to fifth-degree**



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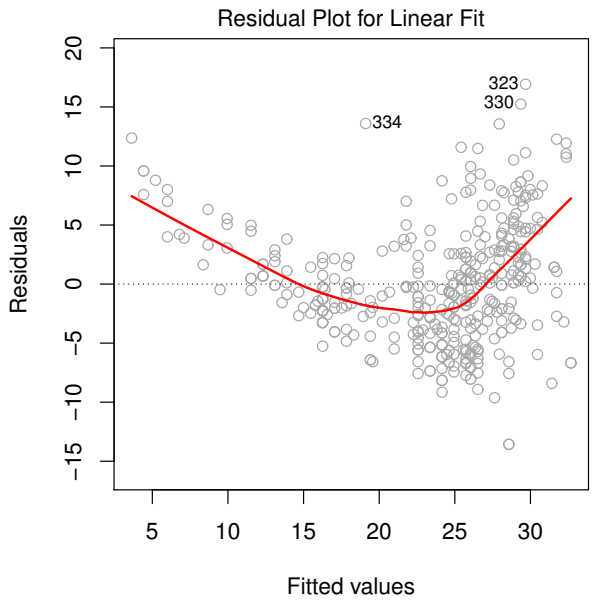
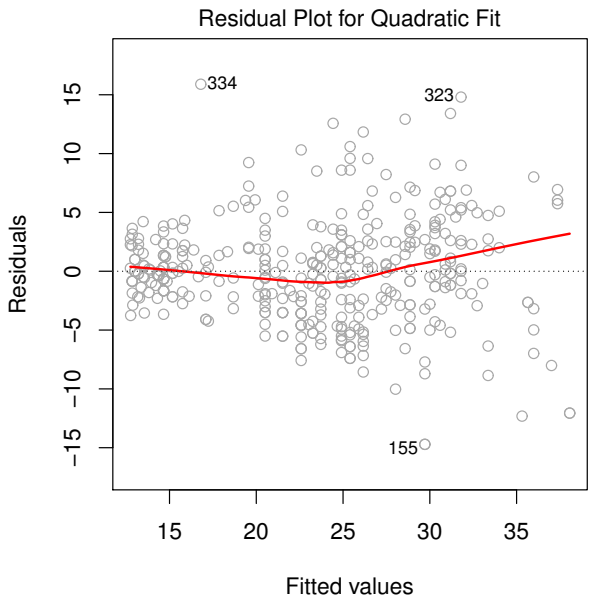
Residual Plots





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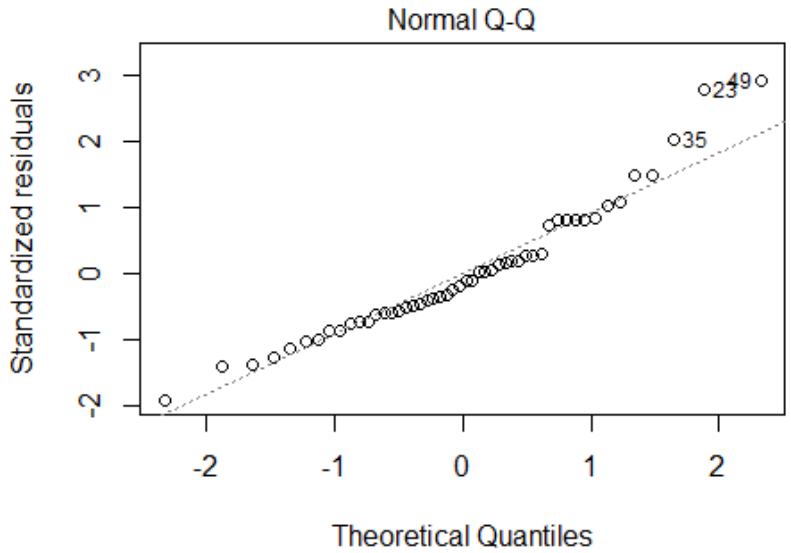
The Normal Probability Plot (QQ Plot)

The best diagnostic test for normally distributed errors is a ***normal probability plot* or Q-Q plot of the residuals:** a plot of the quantiles of residual distribution *versus* the quantiles of a normal distribution having the same mean and variance.

If the distribution is normal, the points on QQ plot should fall close to the diagonal line.

- A ***bow-shaped*** pattern of deviations from the diagonal indicates that the residuals have excessive *skewness* (i.e., they are not symmetrically distributed, with too many large errors in the same direction).
- An ***S-shaped*** pattern of deviations indicates that the residuals have excessive *kurtosis*--i.e., there are either two many or two few large errors in both directions.



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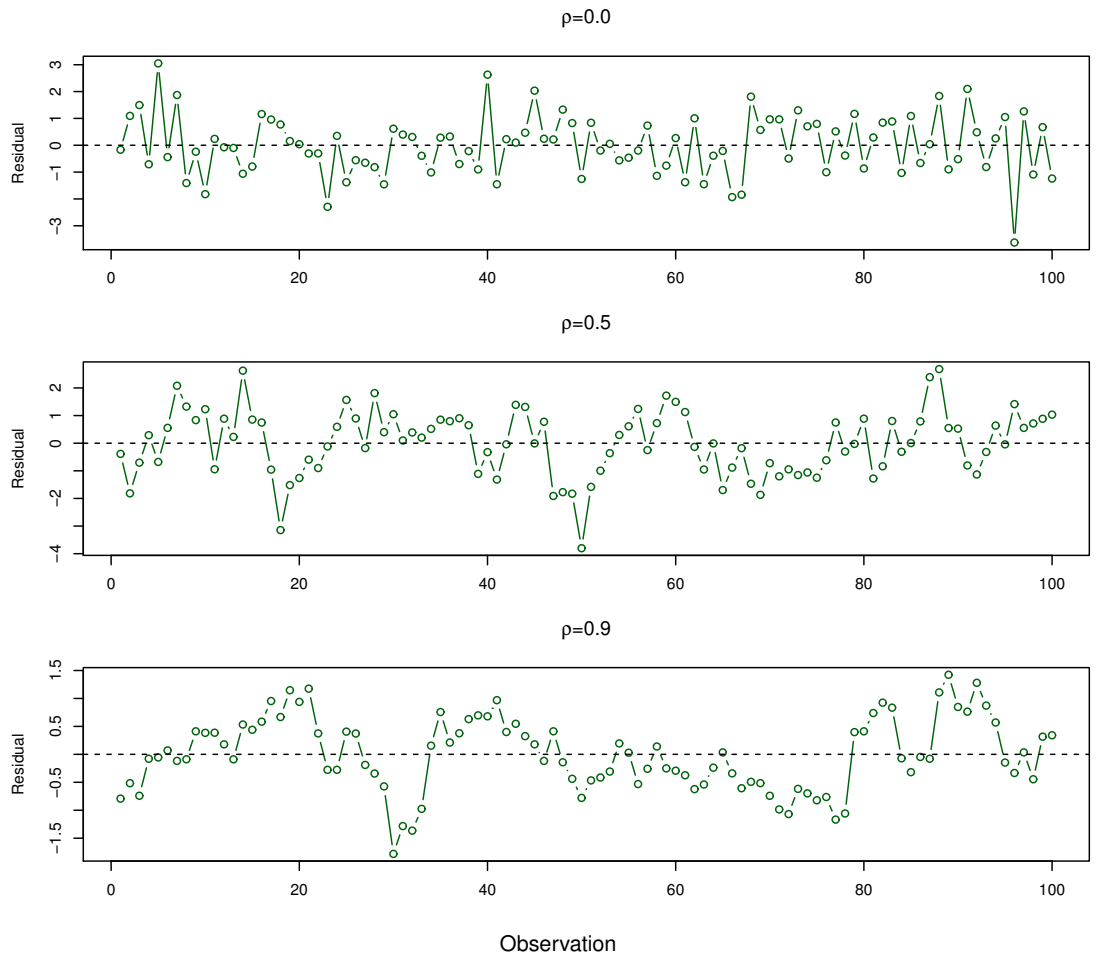
Plots of Residuals from Simulated Time Series Data Sets: the analysis of the first order autocorrelation



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The Durbin-Watson Test

The ***Durbin-Watson statistic*** provides a test for significant residual autocorrelation at lag-1:

$$DW = \frac{\sum_{t=2}^{T}(e_t - e_{t-1})^2}{\sum_{t=1}^{T} e_t^2} \approx 2(1-\hat{\rho})$$

where $\hat{\rho}$ is the lag-1 residual autocorrelation, so ideally it should be close to 2, say, between 1.4 and 2.6 for a sample size of 50.

✓ $DW = 0$ Perfect positive autocorrelation

✓ $DW = 4$ Perfect negative autocorrelation

✓ $DW = 2$ No correlation

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Durbin-Watson's probability table

Tavole per la statistica di Durbin-Watson, livello di signi

n	$k'=1$		$k'=2$		$k'=3$		$k'=4$		
	d_L	d_U	d_L	d_U	d_L	d_U	d_L	d_U	
15	1.077	1.361	0.945	1.543	0.814	1.750	0.685	1.977	0.56
16	1.106	1.371	0.982	1.539	0.857	1.728	0.734	1.935	0.615
17	1.133	1.381	1.015	1.536	0.897	1.710	0.779	1.900	0.664
18	1.158	1.392	1.046	1.535	0.933	1.696	0.820	1.872	0.710
19	1.180	1.401	1.075	1.535	0.967	1.685	0.859	1.848	0.752
20	1.201	1.411	1.100	1.537	0.998	1.676	0.894	1.828	0.792
21	1.221	1.420	1.125	1.538	1.026	1.669	0.927	1.812	0.828
22	1.240	1.429	1.147	1.541	1.053	1.664	0.958	1.797	0.863
23	1.257	1.437	1.168	1.543	1.078	1.660	0.986	1.786	0.895
24	1.273	1.446	1.188	1.546	1.101	1.657	1.013	1.775	0.925
25	1.288	1.454	1.206	1.550	1.123	1.654	1.038	1.767	0.953
26	1.302	1.461	1.224	1.553	1.143	1.652	1.062	1.759	0.979
27	1.316	1.468	1.240	1.556	1.162	1.651	1.083	1.753	1.004
28	1.328	1.476	1.255	1.560	1.181	1.650	1.104	1.747	
29	1.341	1.483	1.270	1.563	1.198	1.650	1.124	1.743	
30	1.352	1.489	1.284	1.567	1.214	1.650	1.143		
	1.363	1.496	1.297	1.570	1.229	1.650	1.160		
		1.502	1.309	1.573	1.244	1.650			

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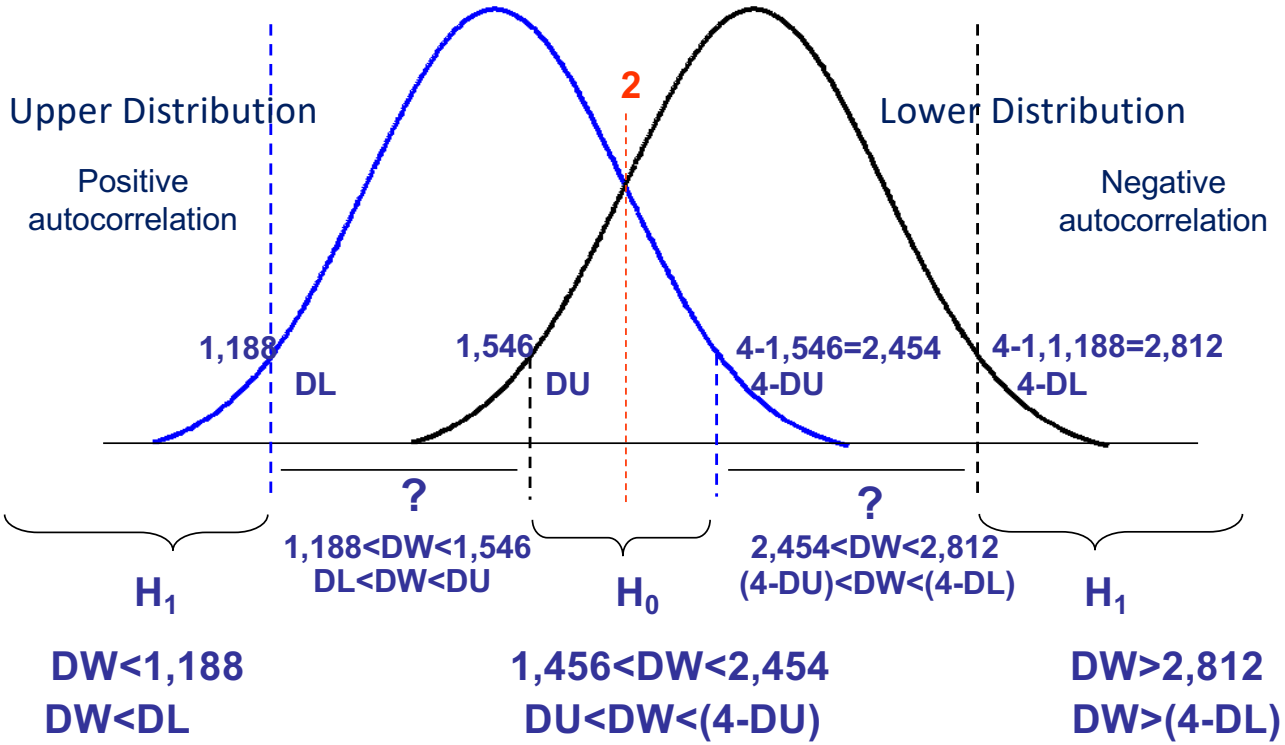
Example of Durbin-Watson's Empirical values

Example: significance 5%, K=2 – plus intercept, n=24



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Heterocedasticy: how to detect

Look at ***plots of residuals versus time*** and ***residuals versus predicted value:***

- ✓ be alert for evidence of *residuals that are getting larger* (i.e., more spread-out) either as a function of time or as a function of the predicted value.

You might also want to plot **residuals *versus* some of the independent variables**:

- ✓ it yields to *confidence intervals that are too wide or too narrow.*

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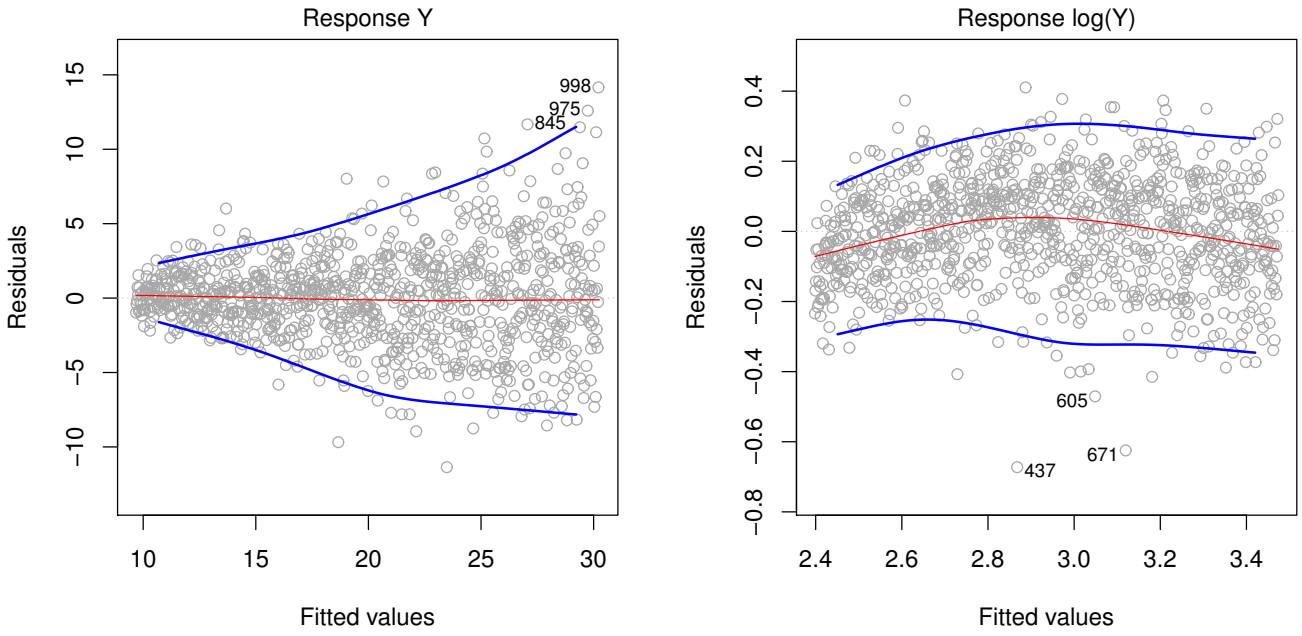

Residual Plots to detect Heteroscedasticity

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Testing homoscedasticy

Bartlett Test and Levene Test

- It tests if the variances of the errors across two samples is equal. It is a $\chi 2$ test.
- Bartlett's test is sensitive to departures from normality.
- The Levene test is less sensitive to departures from normality.

White Test and Breusch Pagan Test

- It tests whether the variance of the errors from a regression is dependent on the values of the independent variables. It is a $\chi 2$ test.
- White Test uses fitted values of the model and one independent variable.
- Breusch Pagan Test using more independent variables performs multiple testing with the the Bonferroni or Holm p-value adjustment.

Score Test and F Test

- The test uses the fitted values of the model, the predictors in the model and a subset of the independent variables. Score Test is a $\chi 2$ test.

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White's Test

Null Hypothesis: $var(\varepsilon_i|\mathbf{X}) = \sigma^2 \ \forall i$

The idea is that $E\left(\varepsilon_i^2|\mathbf{X}\right) = \sigma^2 \ \forall i$ is not correlated with any transformation of the predictors

$$var(\varepsilon_i|\mathbf{X},\mathbf{W}) = f(\mathbf{W},\delta)$$

$$E\left(\varepsilon_i^2|\mathbf{X},\mathbf{W}\right) + u_i$$
$$= f(\mathbf{W},\delta) + u_i \qquad E(u_i|\mathbf{X},\mathbf{W}) = 0$$

Run the regression adding to the set of predictors also a power of the predictors (i.e., the square) and their interactions.

Find the R^2 of the auxiliary model.

Test the significance of $n \times R^2 \sim \chi_g^2$ where g is the number of all regressors in the auxiliary equation model.

Note that the regression of residuals against the predictors can take linear or non-linear form.

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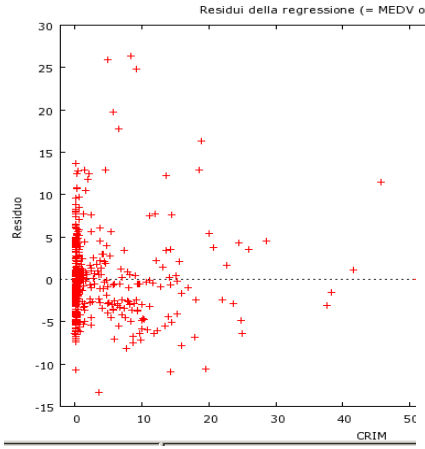

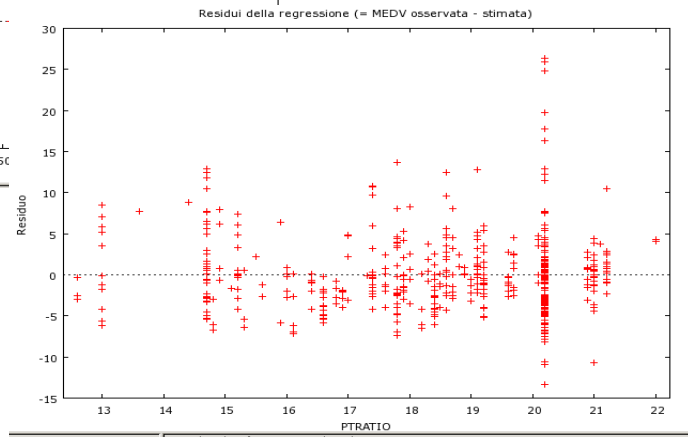

Boston Housing Data – UCI Machine Learning Repository

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Test di White per l'eteroschedasticità -
  Ipotesi nulla: eteroschedasticità non presente
  Statistica test: TR^2 = 308,185
  con p-value = P(Chi-Square(65) > 308,185) = 2,66714e-033
```

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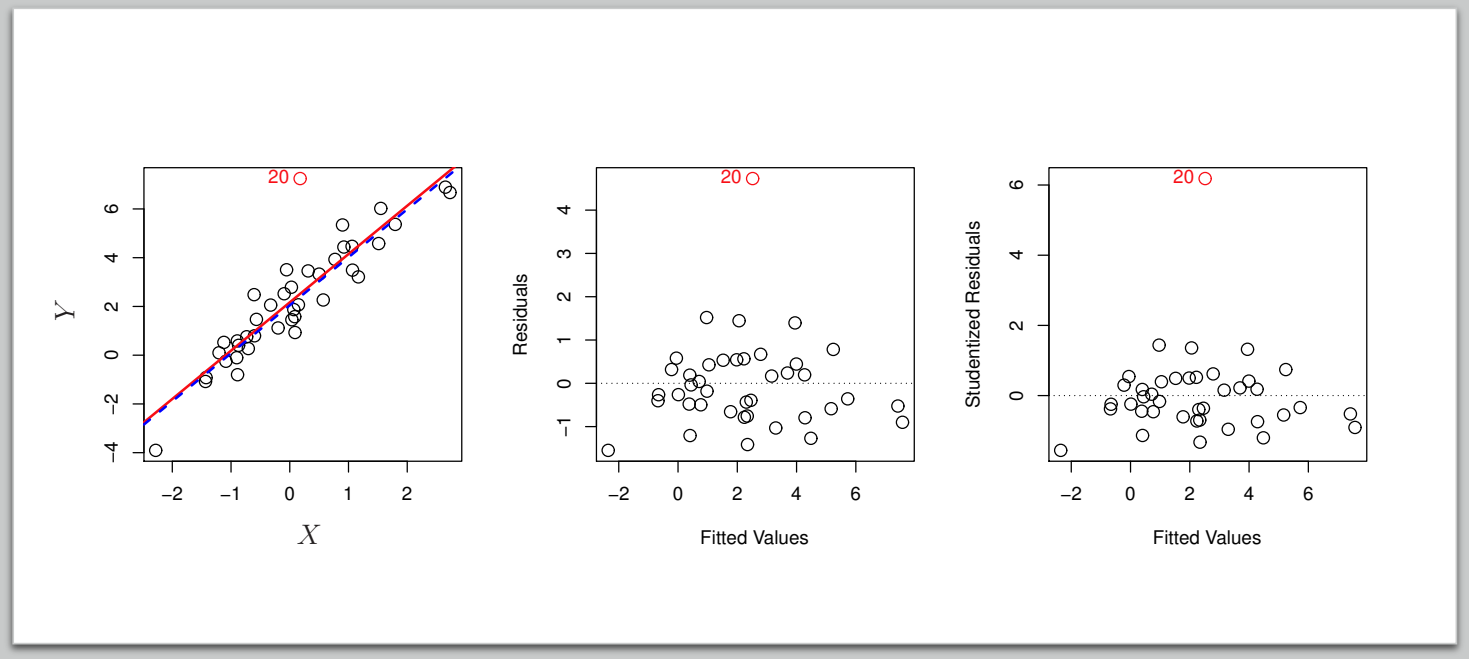

Outliers

An outlier is a point for which y_i is far from the value predicted by the model.

It causes reduction in the amount of explained variation to the fitted line as well as an increase in Residual Sum of Squares, thus reducing the R^2.

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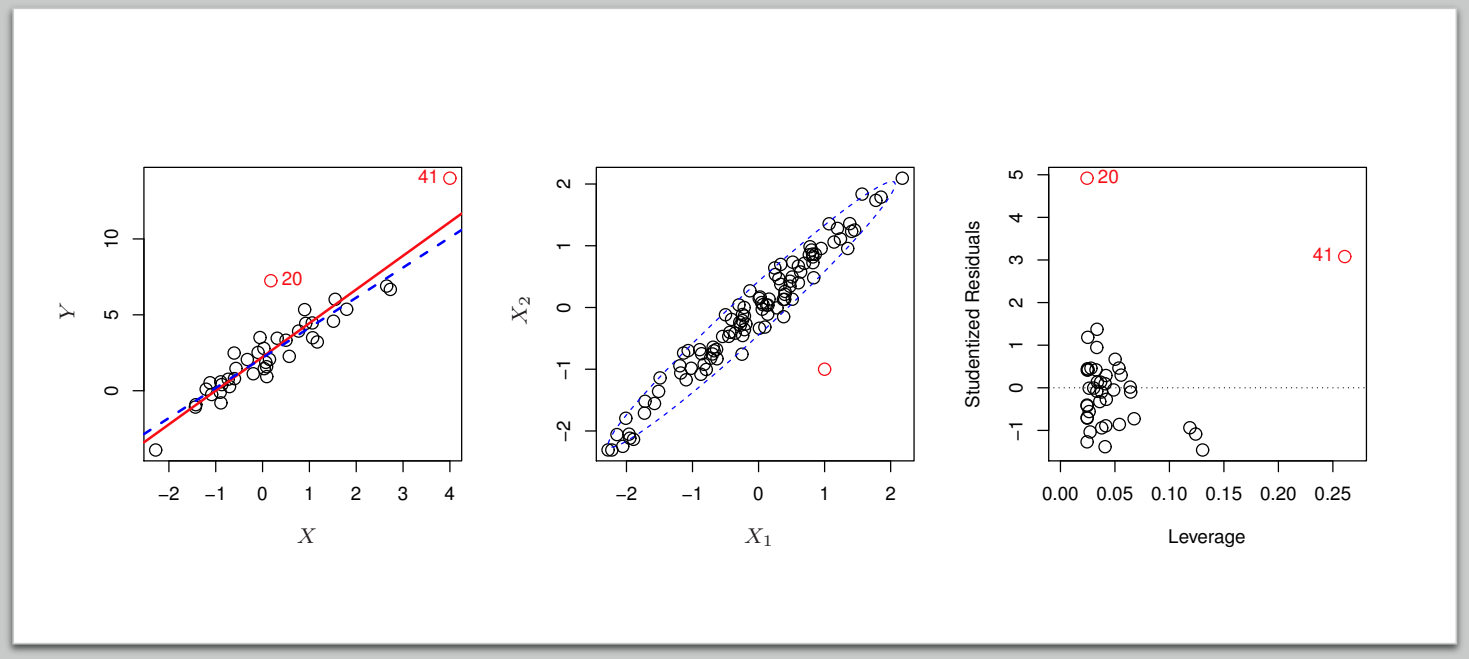

High-Leverage Points

Observations with high-leverage have an unusual value for x_i

The leverage statistic is $h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{l=1}^{n}(x_l - \bar{x})^2}$

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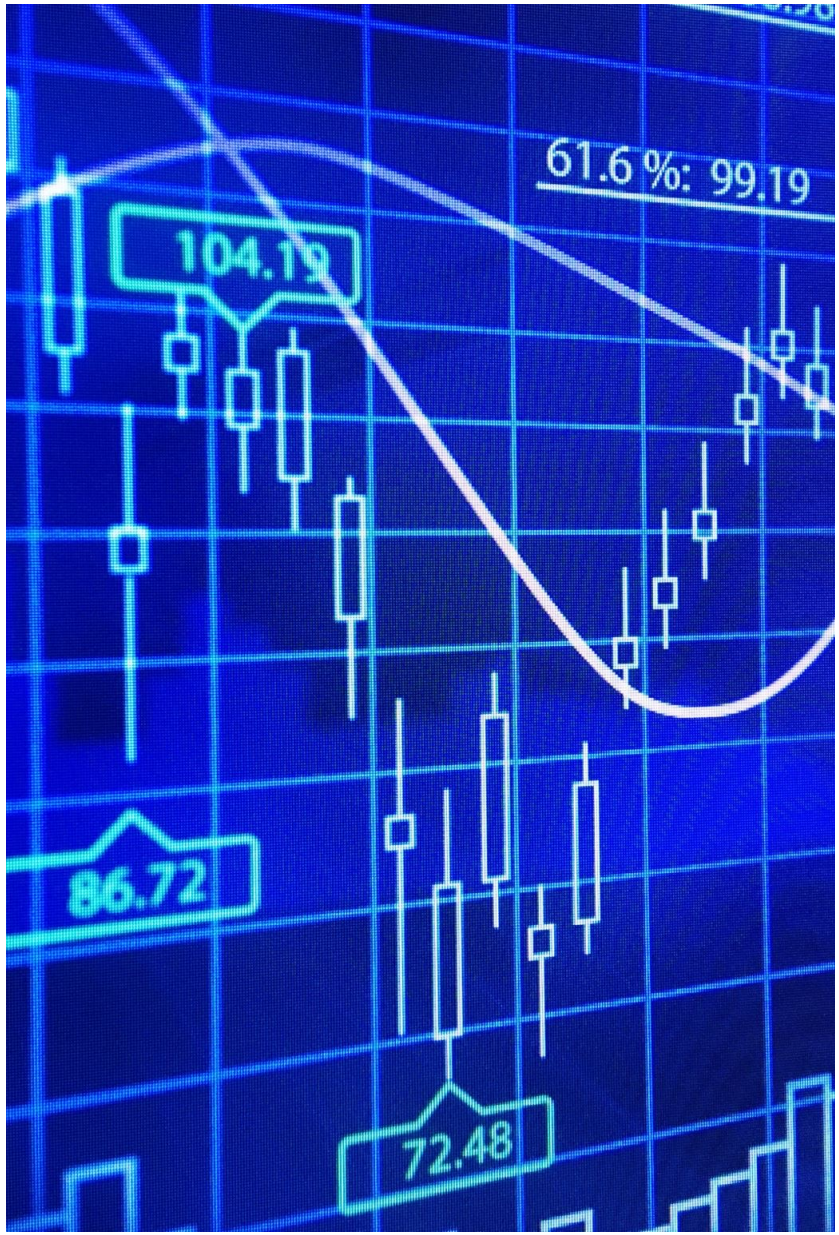

Analysis of Residuals

HAT Matrix or projection matrix $\hat{y}_i = \mathbf{H}y_i = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T y_i$ maps the vector of observed values y_i to the vector of fitted values $\hat{y}_i$.

The diagonal elements are the **leverages** which describe the influence each observed value has on the fitted value.

Cook's Distance:

$$D_i = \frac{\sum_{l=1}^{n}\left(\hat{y}_l - \hat{y}_{l(i)}\right)^2}{p \times \hat{\sigma}^2} = \frac{e_i^2}{p \times \hat{\sigma}^2} \times \left[\frac{h_{ii}}{(1-h_{ii})^2}\right]$$

to identify data points with large residuals (outliers) and/or high leverage, such to reduce accuracy of the regression model.

Recursive Residuals and CUSUM Test for structural change in time series

Studentized residuals $t_i = \frac{\boldsymbol{e_i}}{\hat{\boldsymbol{\sigma}}\sqrt{\mathbf{1}-\boldsymbol{h_{ii}}}}$

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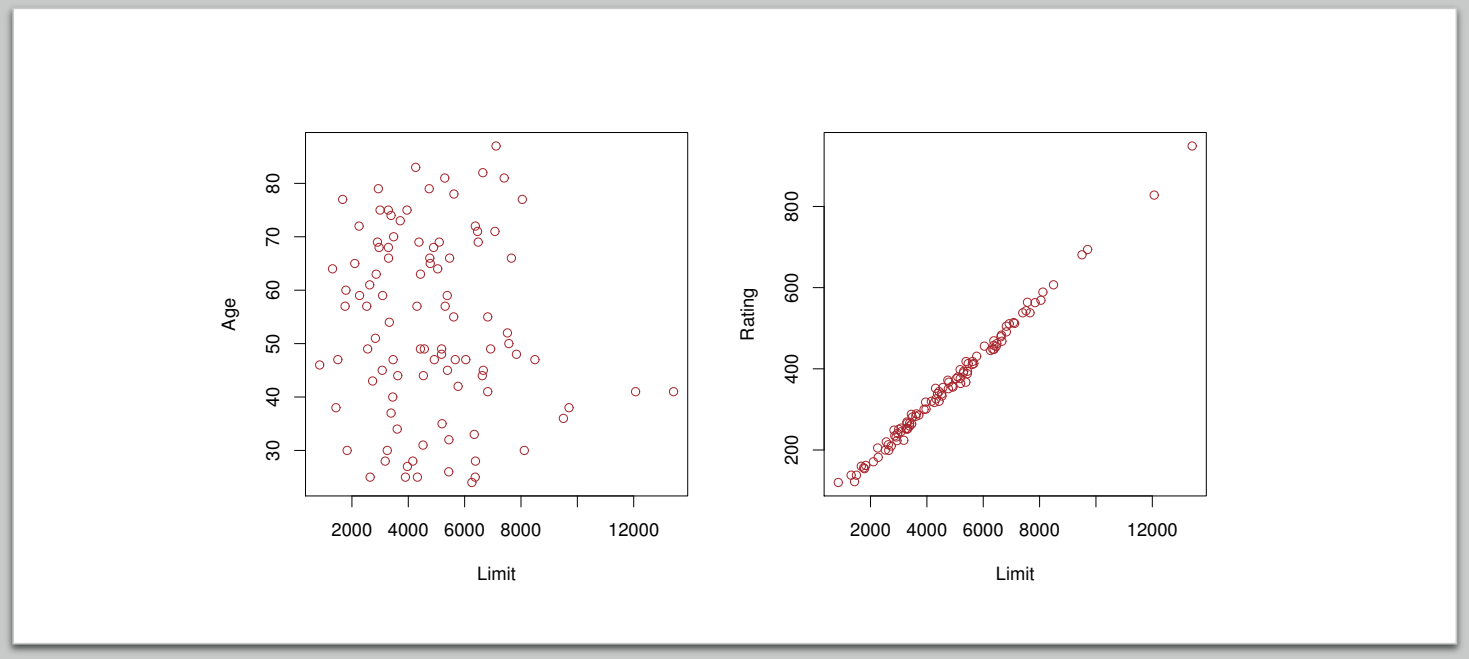

About collinearity

Collinearity refers to the situation in which two or more predictors are closely related to one another.

Credit data set

- **Limit** and **age** appear no correlated
- **Limit** and **rating** are highly correlated

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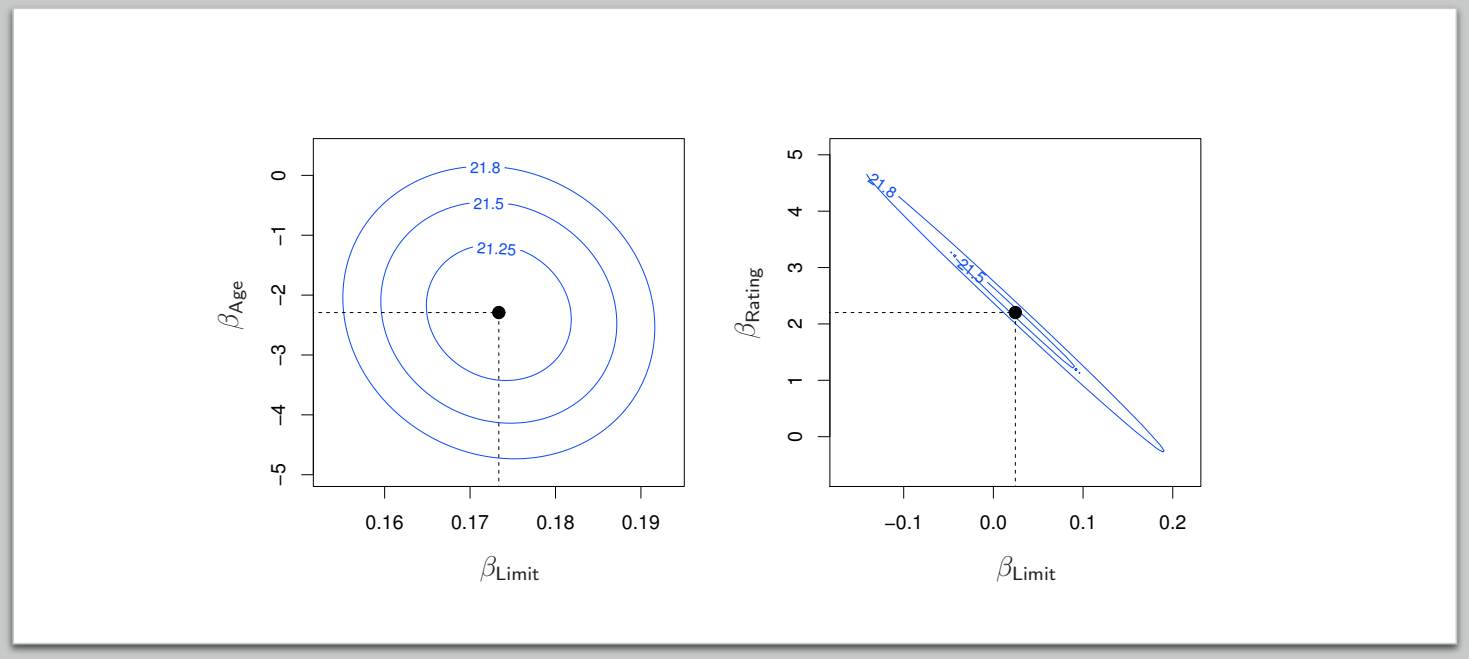

Contour Plots for the RSS values as a function of the parameter estimates

Credit data set

- ✓ **Limit** and **age** appear no correlated as the minimum value is well defined
- ✓ **Limit** and **rating** are highly correlated because there are many pairs $(\beta_{Limit}, \beta_{Rating})$ with a similar value for RSS

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		Coefficient	Std. error	T-statistic	P-value
Model 1	Intercept	-173.411	43.828	-3.957	<0.0001
	`age`	-2.292	0.672	-3.407	0.0007
	`limit`	0.173	0.005	34.496	<0.0001
Model 2	Intercept	-377.537	45.254	-8.343	<0.0001
	`rating`	2.202	0.952	2.312	0.0213
	`limit`	0.025	0.064	0.384	0.7012

Results of two multiple linear regression

Credit data set

- ✓ **Limit** and **age** appear no correlated
- ✓ **Limit** and **rating** are highly correlated as the standard error of $\hat{\beta}_{Limit}$ increases 12-fold in the second regression due to collinearity

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Collinearity

How to detect: high R^2 value with un-significant regressors, high simple correlations, high values of VIF (Variance Inflation Factors) computed for each predictor at turn

$$VIF(\hat{\beta}_j) = \frac{1}{1 - R^2_{X_j|X_{-j}}}$$

Where $R^2_{X_j|X_{-j}}$ is the R^2 from a regression of X_j onto all of the other predictors. If $R^2_{X_j|X_{-j}}$ is close to one, then collinearity is present, and so the $VIF(\hat{\beta}_j)$ will be large.

How to fix: skip those predictors that are highly correlated to the others.

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